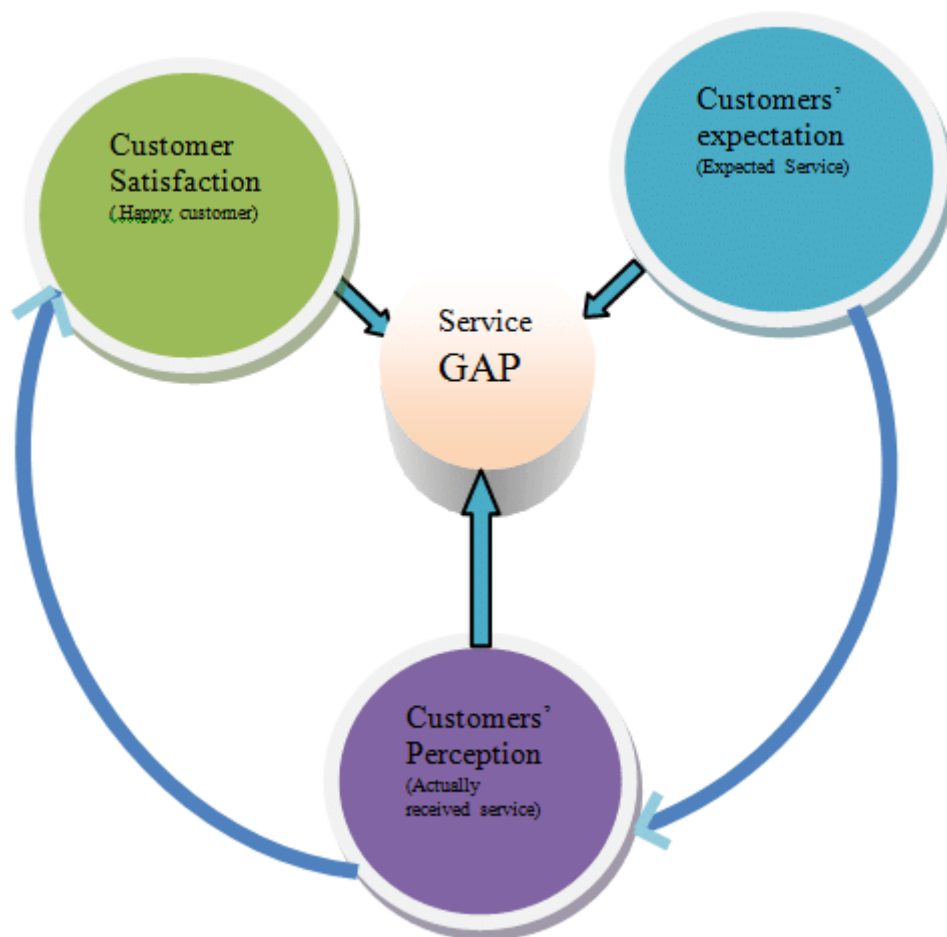


UNIT – II

CUSTOMER SATISFACTION, EMPLOYEE INVOLVEMENT AND CONTINUOUS IMPROVEMENT

2.1 Customer Satisfaction

Customer satisfaction is a business concept that refers to the degree to which products and services provided by an organization meet or exceed customer expectations. In modern quality management systems, customer satisfaction is considered a **key performance indicator (KPI)** and forms one of the four perspectives of the **Balanced Scorecard**. Organizations operating in competitive markets increasingly treat customer satisfaction as a strategic requirement rather than a supporting activity.



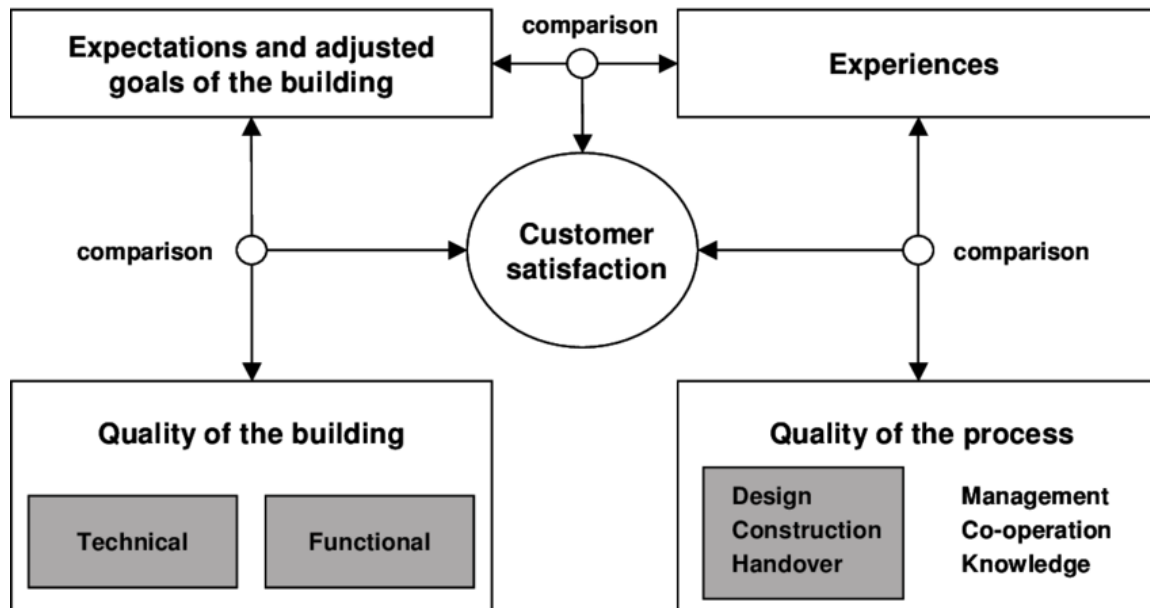


Figure 2.1 Customer Satisfaction Concept Model

Customer satisfaction is influenced by several key factors such as **performance, features, service, warranty, price, and reputation**. When customers perceive that the value received from a product or service is greater than or equal to their expectations, satisfaction is achieved. Consistently satisfied customers are more likely to remain loyal and contribute to positive word-of-mouth.

2.1.1 Customer Complaints and Customer Categories

Customer feedback generally places customers into three categories: **satisfied customers, dissatisfied customers, and totally dissatisfied customers**. Satisfied customers continue to support the business, while dissatisfied customers may complain or switch to competitors. Totally dissatisfied customers not only stop purchasing but may also negatively influence others. Therefore, customer complaints must be systematically monitored and analyzed as they serve as early warning signals of quality and service issues.

2.2 Feedback and Voice of the Customer

Feedback is the process of collecting information from customers to evaluate satisfaction levels and identify improvement opportunities. Effective feedback systems help organizations discover customer satisfaction, determine quality priorities, compare performance with competitors, identify customer needs, and locate opportunities for continuous improvement.

2.2.1 Customer Feedback Tools

Customer feedback can be collected using various tools such as comment cards, customer questionnaires, focus groups, toll-free telephone numbers, customer visits, report cards, internet-based feedback systems, employee feedback, and mass customization data.

Customer Questionnaire and Rating Scale

Customer questionnaires often use structured rating scales to quantify satisfaction levels. A typical five-point scale ranges from **highly satisfied** to **highly dissatisfied** and allows organizations to compare responses across different service attributes.

Maintenance Staff

Based on your most recent maintenance work order, please rate the following aspects of your maintenance staff .

	Poor	Fair	Average	Good	Excellent	N/A
Overall satisfaction with maintenance	☐	☐	☐	☐	☐	☐
Professionalism	☐	☐	☐	☐	☐	☐
Quality of work	☐	☐	☐	☐	☐	☐
Timeliness of completion of work	☐	☐	☐	☐	☐	☐
Notification of completed work	☐	☐	☐	☐	☐	☐
Cleanliness after work performed	☐	☐	☐	☐	☐	☐

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
I am satisfied with the quality of taste	☐	☐	☐	☐	☐
I am satisfied with the price	☐	☐	☐	☐	☐
I am satisfied with the appearance	☐	☐	☐	☐	☐
I am satisfied with the co-operation given	☐	☐	☐	☐	☐
I am satisfied with the sales service	☐	☐	☐	☐	☐

Brief comment

Thank you

Surveys become more effective when organizations recognize that customers and clients may not be the same, ask specific questions, limit survey duration, carefully select respondents, and decide in advance how collected data will be analyzed and used.

2.3 Service Quality

Customer service refers to the set of activities an organization performs to win and retain customer satisfaction. Service quality can be delivered before the sale, during the sale, or after the sale of a product, and in some cases, service itself is the primary offering.

High service quality depends on how well the organization structures its operations and interactions. Organizational service quality begins with identifying market segments, documenting customer requirements, communicating those requirements internally, organizing processes, and arranging physical spaces to support service delivery.

Customer care involves meeting expectations, understanding the customer's viewpoint, delivering promised service, responding promptly to complaints, and making customers feel valued. Effective communication balances time and personal attention, minimizes unnecessary contact points, and ensures that all documents and interactions are customer-friendly.

2.4 Customer Retention

Customer retention refers to the organization's ability to retain customers over time and is more powerful than customer satisfaction alone. Retention requires both **customer satisfaction** and **customer loyalty**. Loyal customers provide stable revenue, reduce marketing costs, and support long-term business growth.

Customer retention improves when top management commits to customer satisfaction, customer preferences are clearly understood, service standards are established, employees are properly trained and rewarded, customer relationships are maintained continuously, and service quality is improved on an ongoing basis.

2.5 Employee Involvement

Employee involvement is a management approach that improves quality and productivity by involving employees in decisions that affect their work. It enables employees to contribute their knowledge and skills, leading to better decision-making, faster problem identification, and improved morale.

2.5.1 Employee Expectations versus Management Perceptions

Studies show that employees value **interesting work**, **appreciation**, and **involvement** more than monetary rewards alone, while managers often assume pay is the primary motivator. Understanding this difference is essential for designing effective motivation and involvement systems.

What Motivates Employees to Go the Extra Mile?



What employees say matter most to motivating performance



2.6 Motivation and Workforce Development

A motivated workforce is developed indirectly through leadership behavior and organizational culture. Motivation improves when leaders understand themselves and their employees, maintain a positive attitude, share organizational goals, monitor progress, and create interesting work through job rotation, job enlargement, and job enrichment.

The **seven rules of motivation** emphasize setting meaningful goals, completing tasks, associating with motivated individuals, developing self-learning skills, aligning talent with interest, continuously increasing knowledge, and accepting risk and failure as learning opportunities.

2.7 Employee Surveys

Employee surveys help organizations assess employee satisfaction, identify trends, measure the effectiveness of quality initiatives, and improve communication. The survey process typically involves forming a multifunctional team through the quality council, designing the survey instrument, administering the survey, analyzing results, and identifying improvement actions.

Employee involvement in decision-making can be structured at different levels such as **Tell, Sell, Consult, Join, and Delegate**, depending on the degree of authority shared with employees.

2.8 Empowerment

Empowerment means providing employees with authority and responsibility to make decisions related to their work. The purpose of empowerment is to utilize the full potential of employees by equipping them with skills and then allowing them to act independently.

Empowerment is based on principles such as defining responsibilities, granting authority, setting performance standards, providing training and information, trusting employees, allowing controlled mistakes, and treating employees with dignity. The three dimensions of empowerment are **capability, alignment, and trust**.

2.9 Teams and Teamwork

A team is a group of individuals working together to achieve common objectives. Teamwork occurs when members place team goals above individual interests. Teams work effectively because collective knowledge is greater, communication improves, and cooperation increases.

Interdependence of the key factors of teamwork



Meeting Guidelines for Teams



Organizations use different types of teams such as process improvement teams, cross-functional teams, natural work teams, and self-directed teams. Successful teams require clear objectives, proper leadership, accountability, trust, open communication, and adequate resources.

2.10 Recognition, Reward and Performance Appraisal

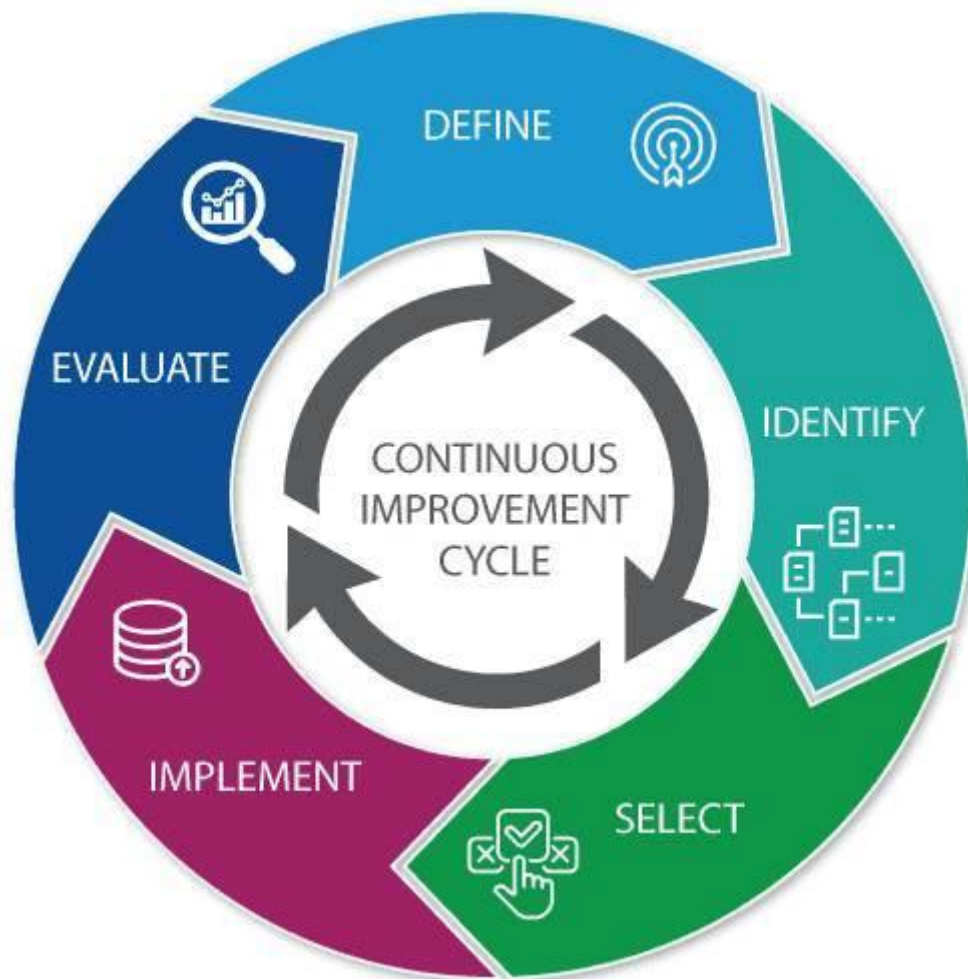
Recognition is the process by which management acknowledges employee performance and contributions. Recognition may take the form of praise, certificates, rewards, promotions, or appreciation letters. Recognition should be timely, while rewards should match the level of contribution.

Recognition and reward systems improve employee morale, encourage involvement, motivate continuous improvement, and demonstrate that the organization values quality and productivity.

Performance appraisal is used to inform employees about their performance and support decisions related to promotion, salary increments, and counseling. Effective appraisal systems focus on fairness, continuous feedback, employee participation, and development rather than punishment.

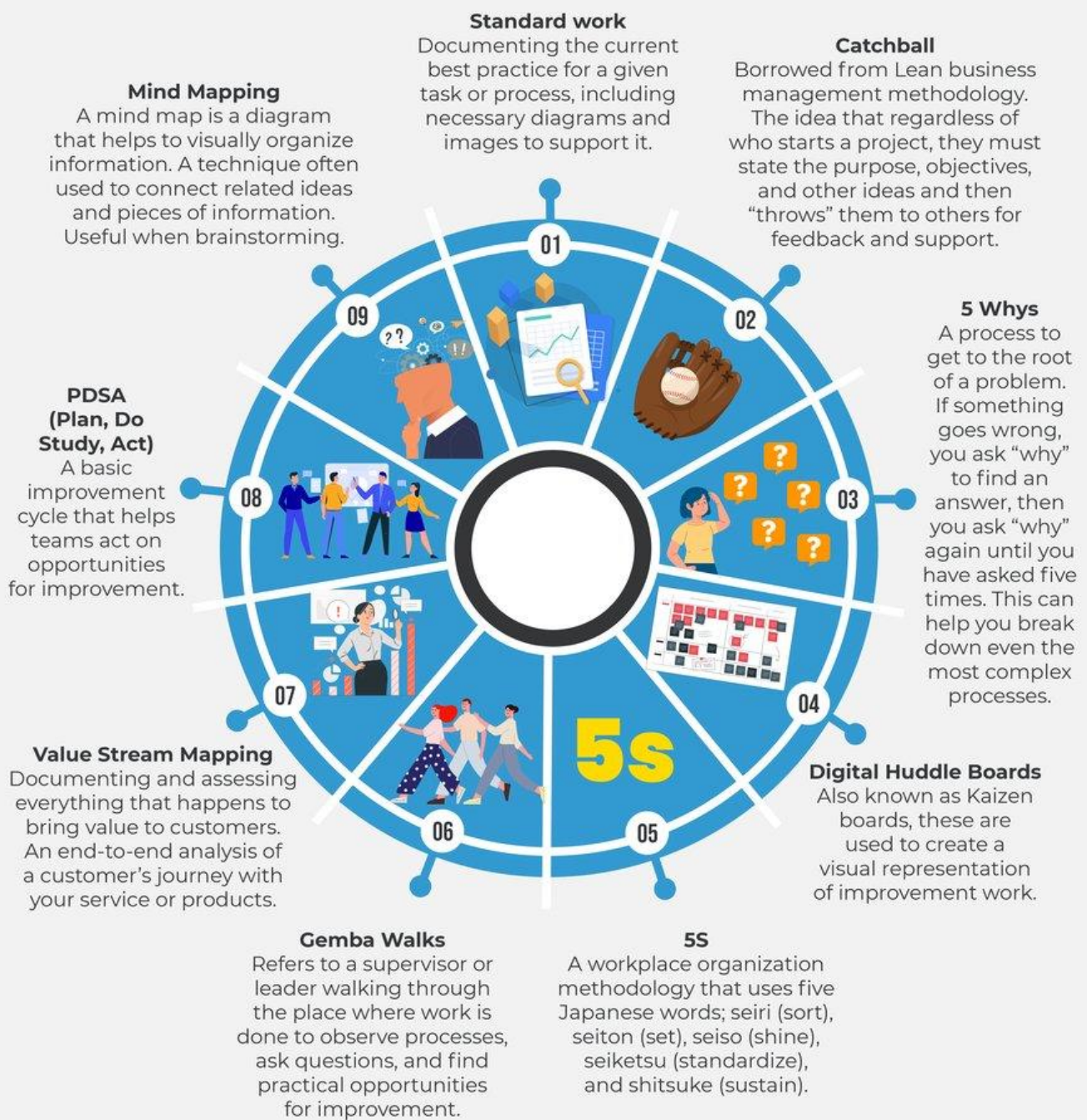
2.11 Continuous Process Improvement

Continuous Process Improvement (CPI) is a structured approach that utilizes organizational resources to build a quality-driven culture. CPI treats all work as a process and focuses on making processes effective, efficient, and adaptable.



CONTINUOUS QUALITY IMPROVEMENT TECHNIQUES

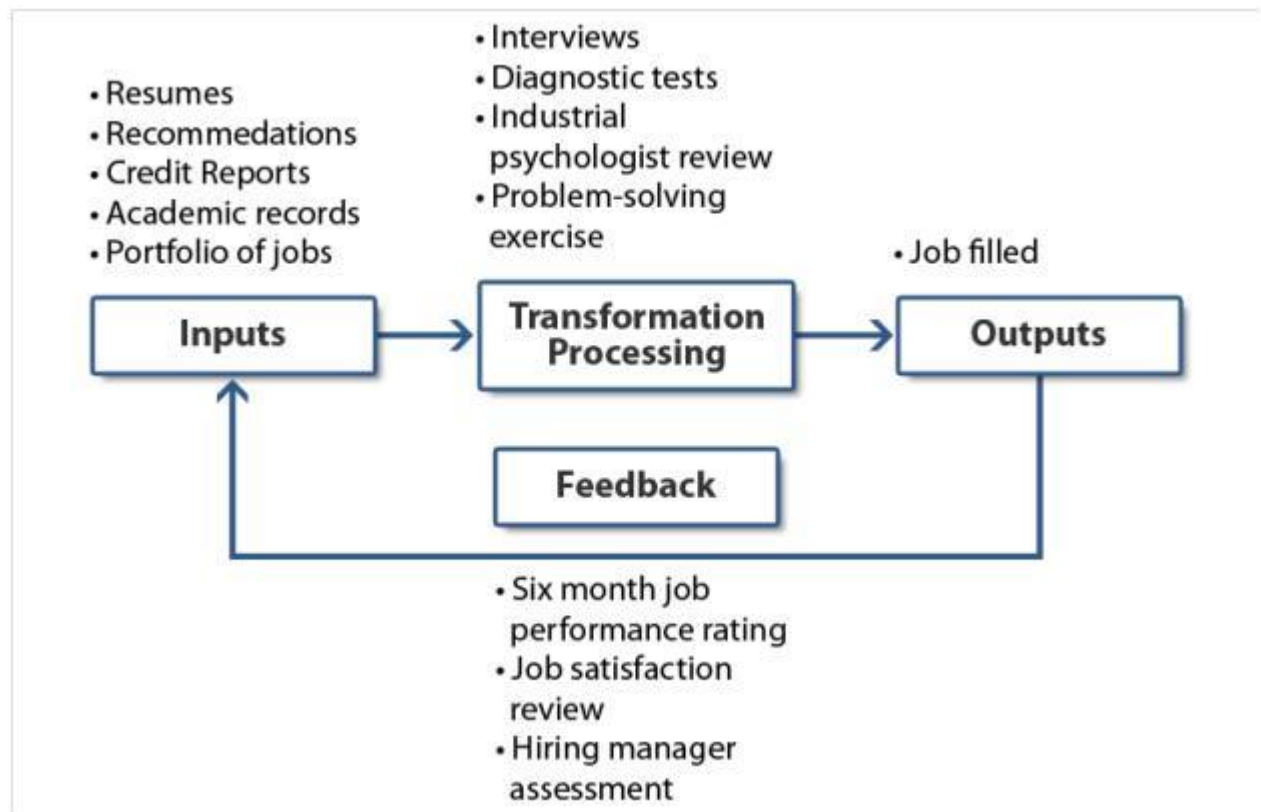
Continuous quality improvement is based on the idea that some of the most effective and impactful things are achieved through simple means.

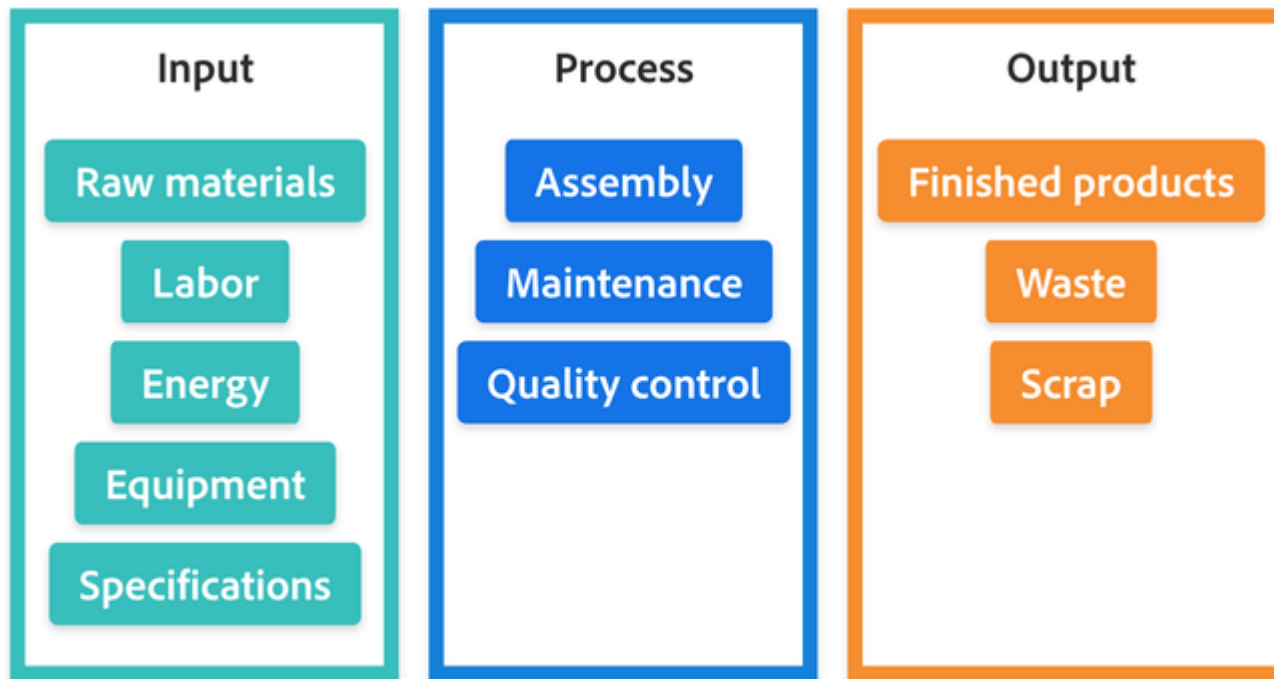


Improvement is achieved by reducing errors, eliminating waste and rework, removing non-value-added activities, controlling performance using statistical tools, applying benchmarking, and incorporating lessons learned into future activities.

2.11.1 Input–Process–Output (IPO) Model

Every process consists of inputs, transformation activities, and outputs. Inputs such as materials, information, and manpower are transformed into outputs through processes, and feedback is used to control and improve performance.



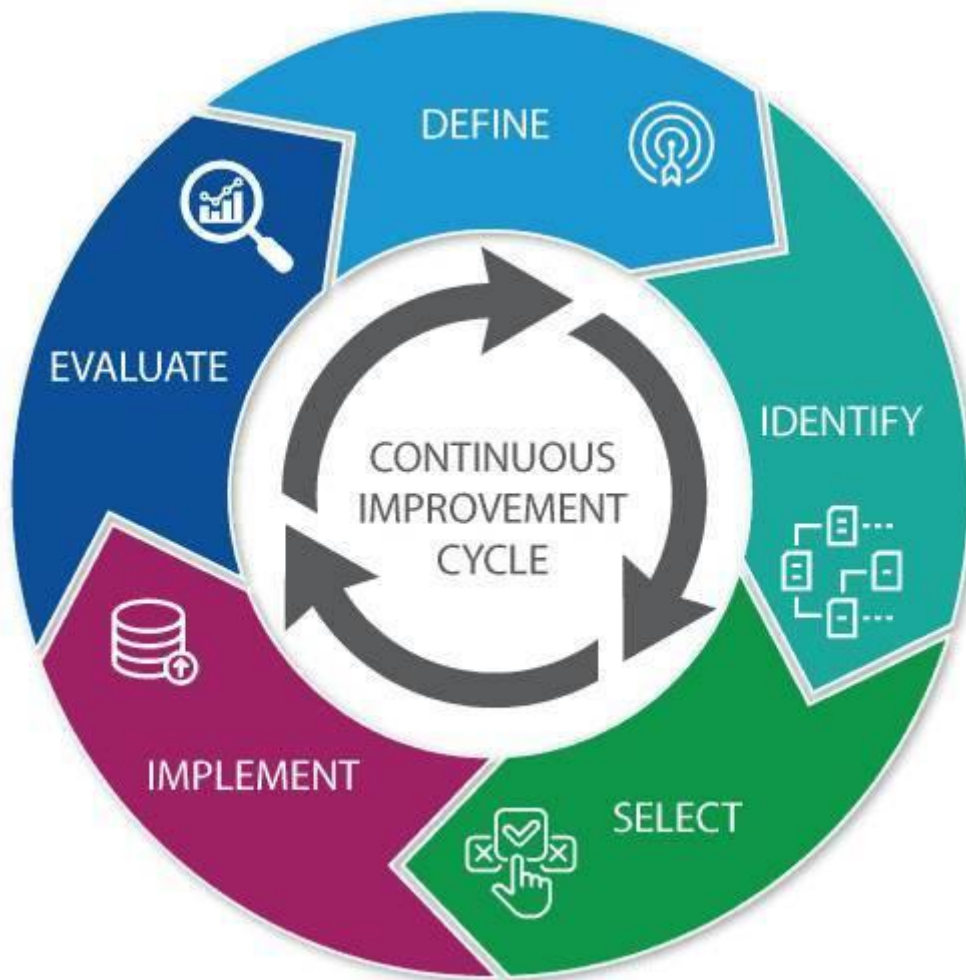


2.11.2 The Juran Trilogy

The Juran Trilogy explains quality management through three managerial processes: **quality planning**, **quality control**, and **quality improvement**. Planning focuses on identifying customer needs and designing processes. Control ensures performance meets standards. Improvement aims at achieving higher performance levels through structured projects.

2.10.3 Continuous Process Improvement Cycle

Continuous process improvement is a structured approach used to improve quality, cost, delivery, and safety by making incremental and sustained improvements. Improvement is achieved by viewing all work as a process, making processes efficient and adaptable, controlling performance using data, eliminating waste and rework, removing non-value-added activities, and applying quality tools such as Statistical Process Control (SPC), benchmarking, experimental design, and Quality Function Deployment (QFD).



This is a visualization of the Deming Cycle

Deming Cycle

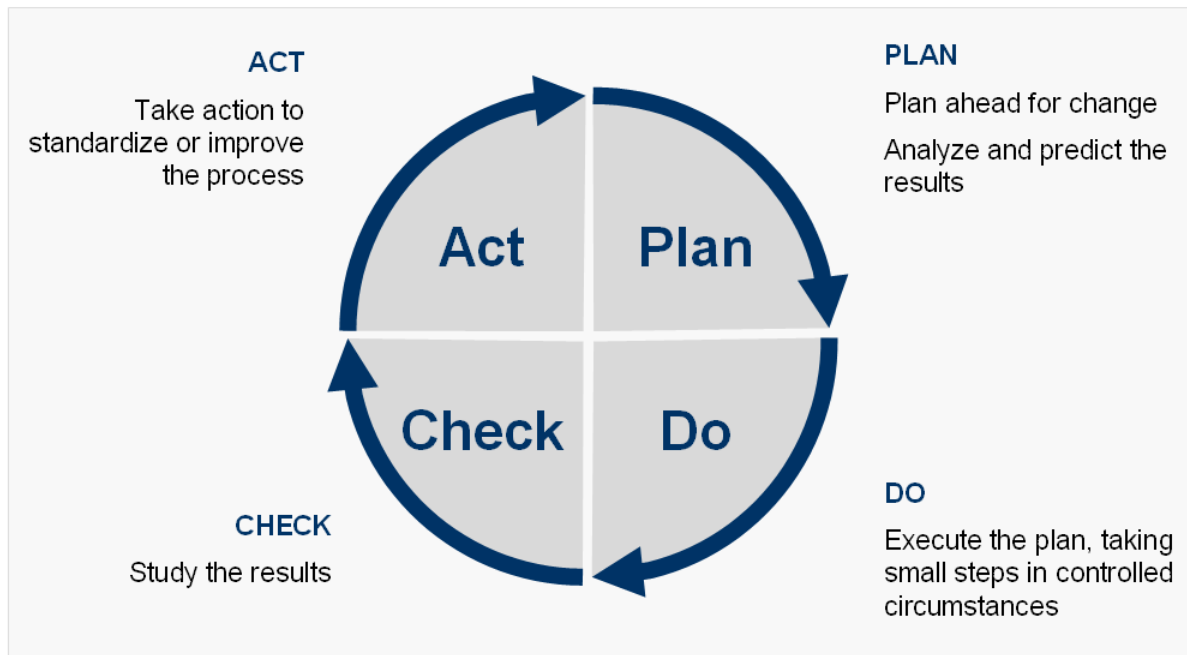
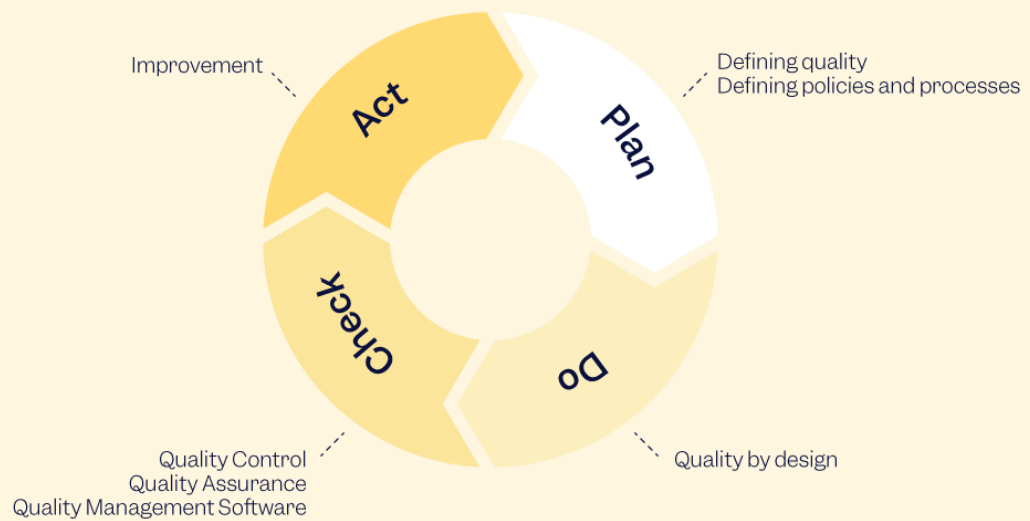


Figure 2.3 Continuous Process Improvement Cycle

2.12 PDCA Cycle (Plan–Do–Check–Act)

The **PDCA cycle** is a systematic improvement methodology used to solve problems and improve processes. In the **Plan** stage, problems are identified and solutions are planned. In the **Do** stage, solutions are implemented on a small scale. In the **Check** stage, results are evaluated using data. In the **Act** stage, successful solutions are standardized or corrective actions are taken.

Continuous improvement in Life Sciences PDCA cycle



Scilife

Plan, do, check, act

Identify the issue

Define the problem, gather relevant data, and formulate a hypothesis. What exactly are you trying to achieve? What resources do you need?

Launch or adjust

If your solution proved effective, how can you implement it more widely? If it fell short, what adjustments are needed? Perhaps you need to return to the planning stage with new insights.

Quickly try out a solution

Implement your plan – but start small. This stage is a controlled experiment, not full-scale deployment.

See if it works

Step back and examine what happened. Did you get the desired result? Why or why not? Compare your actual outcomes against your predictions, looking for insights and unexpected consequences.



BiteSize Learning

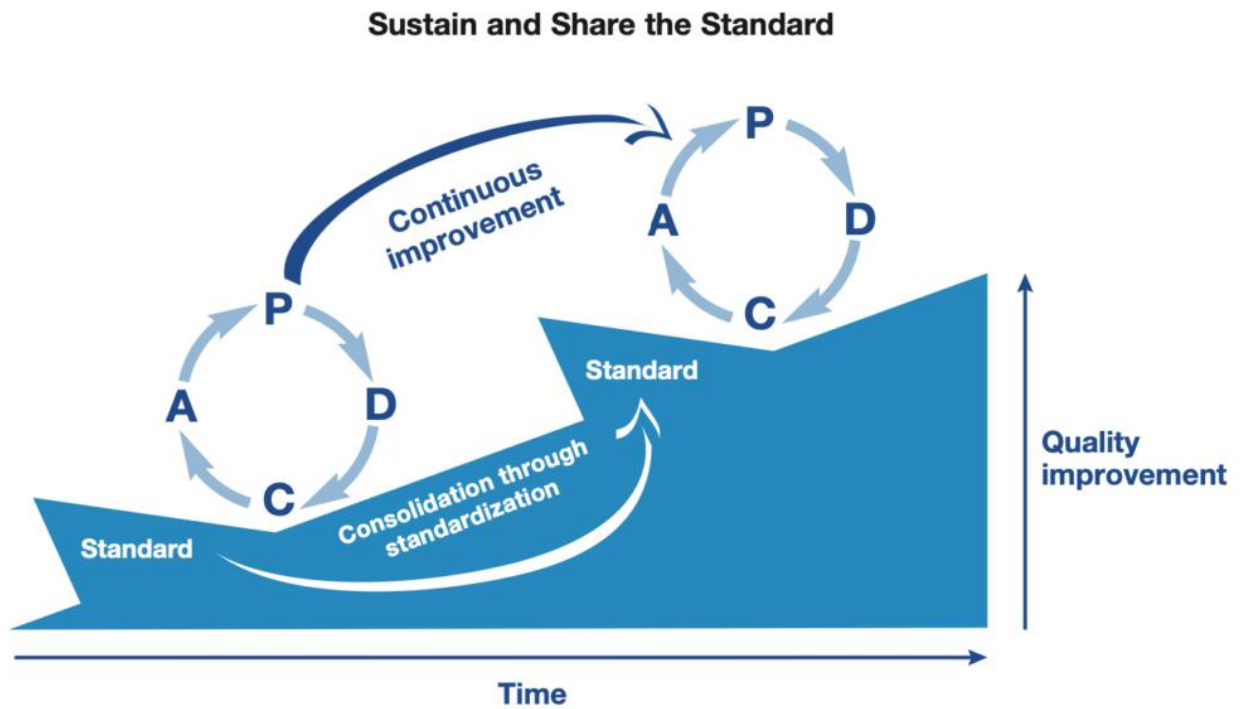
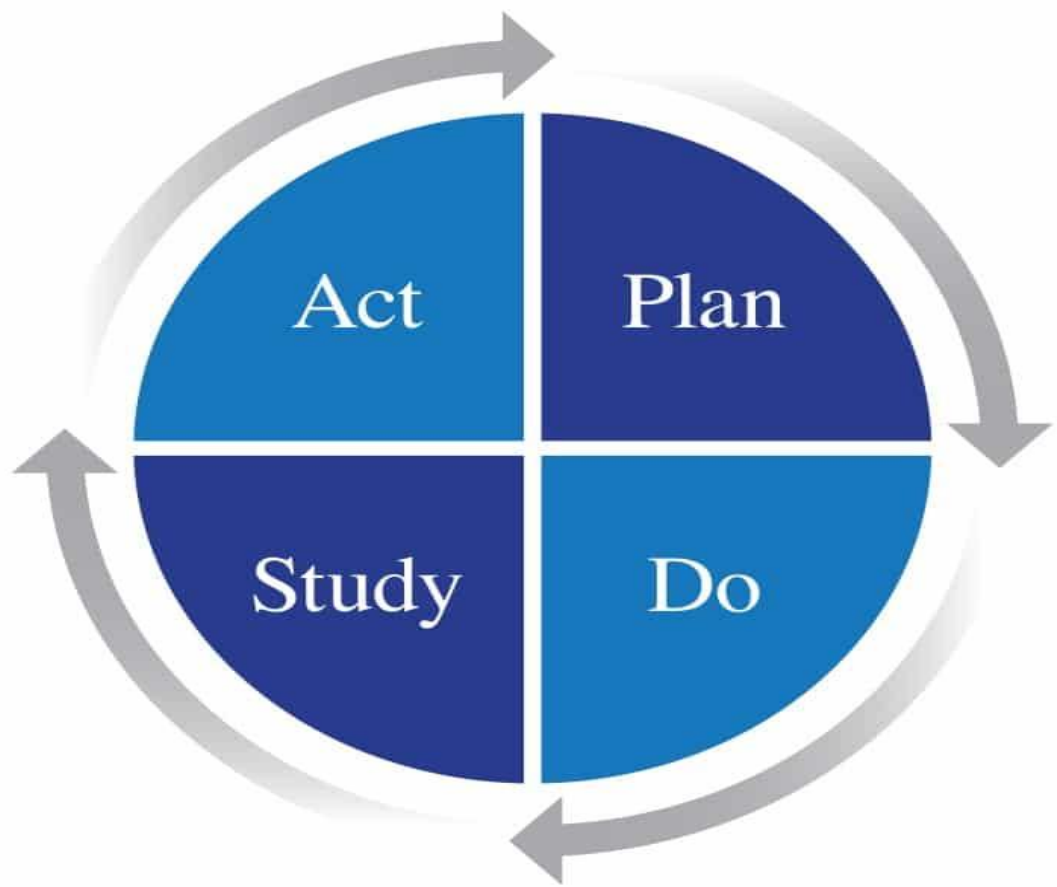


Figure 2.4 PDCA Cycle

2.13 PDSA Cycle (Plan–Do–Study–Act)

The **PDSA cycle** is an enhanced version of PDCA where the emphasis is on **Study** rather than just checking results. Studying helps organizations understand why an improvement worked or failed and supports evidence-based learning.





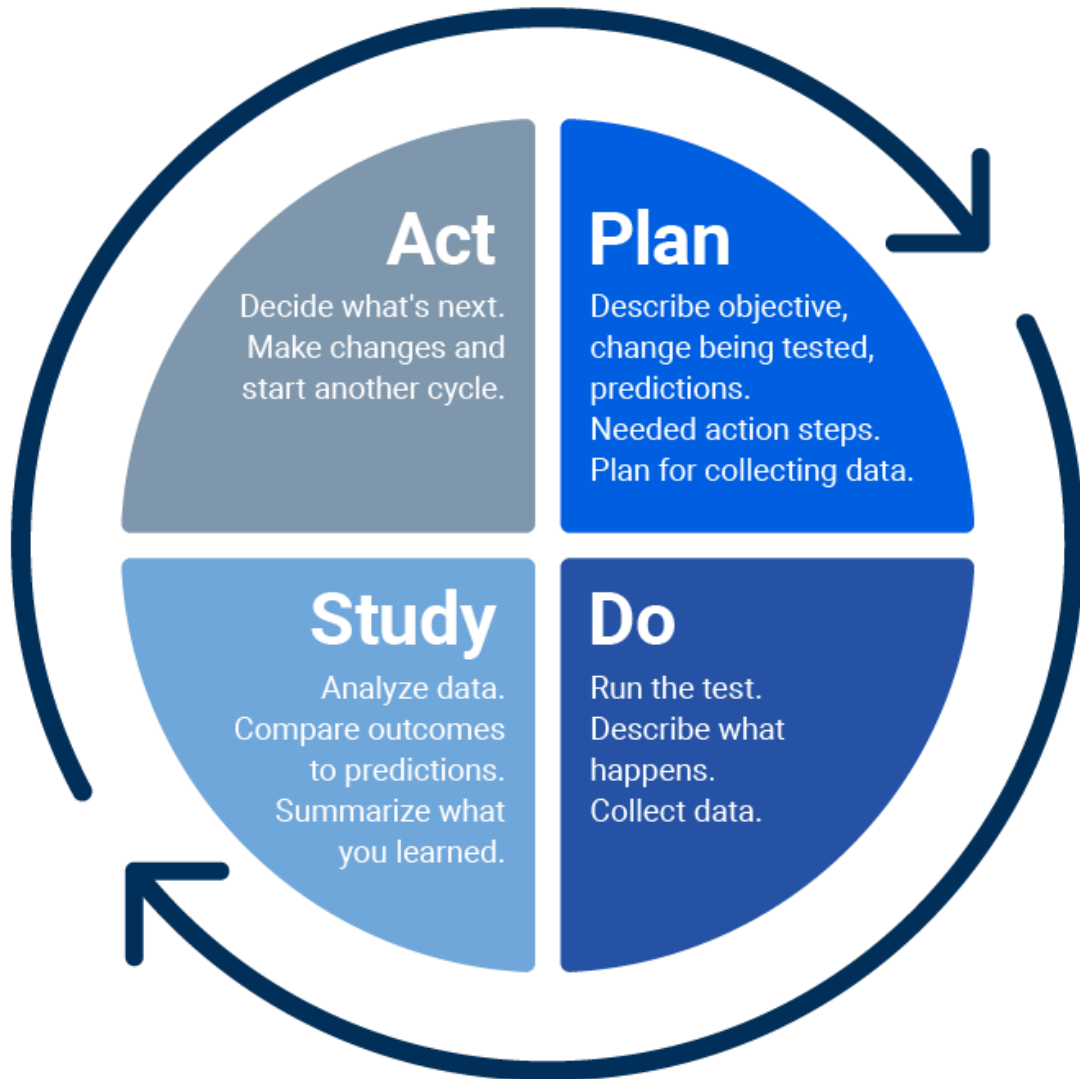
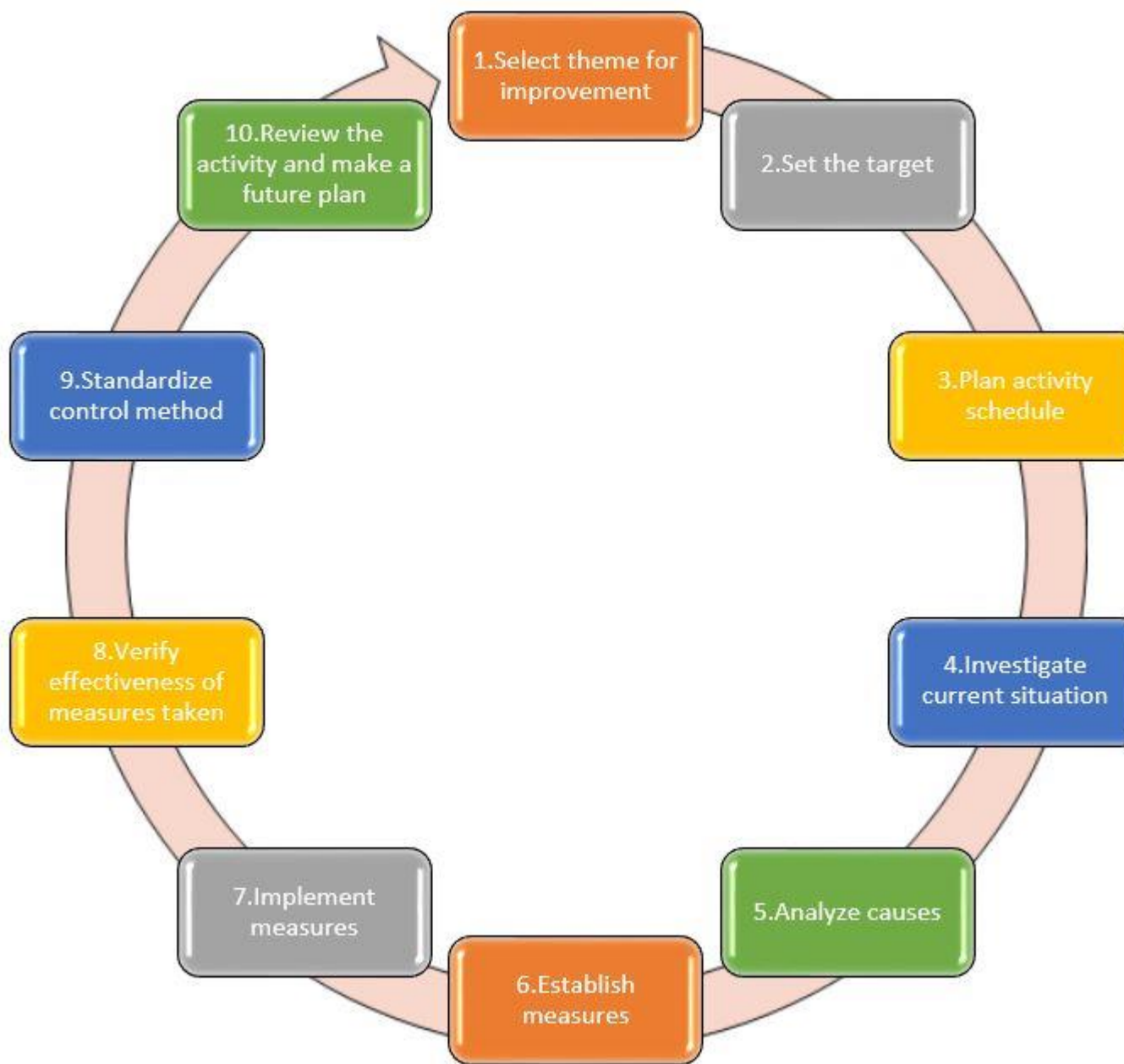


Figure 2.5 PDSA Cycle

2.14 Problem Solving Method (Seven-Step Model)

Problem solving in TQM follows a structured seven-step approach. The process begins with identifying improvement opportunities using customer feedback, Pareto analysis, internal alarms, and brainstorming. The current process is analyzed using flow diagrams and data collection. Optimal solutions are developed using creativity techniques. Approved changes are implemented, results are studied, successful solutions are standardized, and future improvement opportunities are planned.





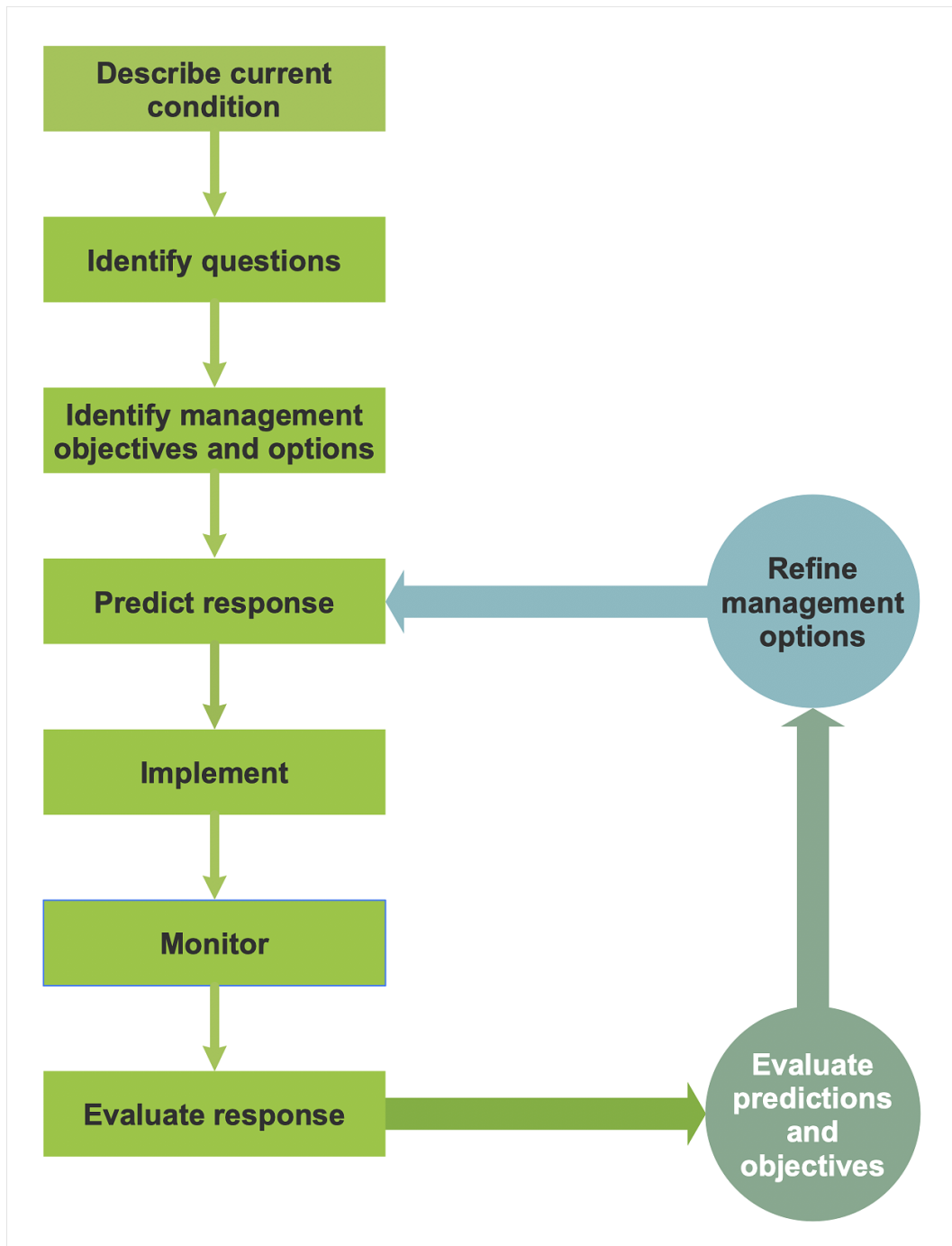


Figure 2.6 Seven-Step Problem Solving Model

2.15 5S Housekeeping System

The **5S system** is a workplace organization method essential for Total Quality Management. It ensures a clean, safe, and efficient workplace, which is necessary for producing consistent quality output.

- **Seiri (Sort)** – Remove unnecessary items

- **Seiton (Set in order)** – Arrange items for easy access
- **Seiso (Shine)** – Clean the workplace
- **Seiketsu (Standardize)** – Maintain cleanliness standards
- **Shitsuke (Sustain/Discipline)** – Follow standards continuously



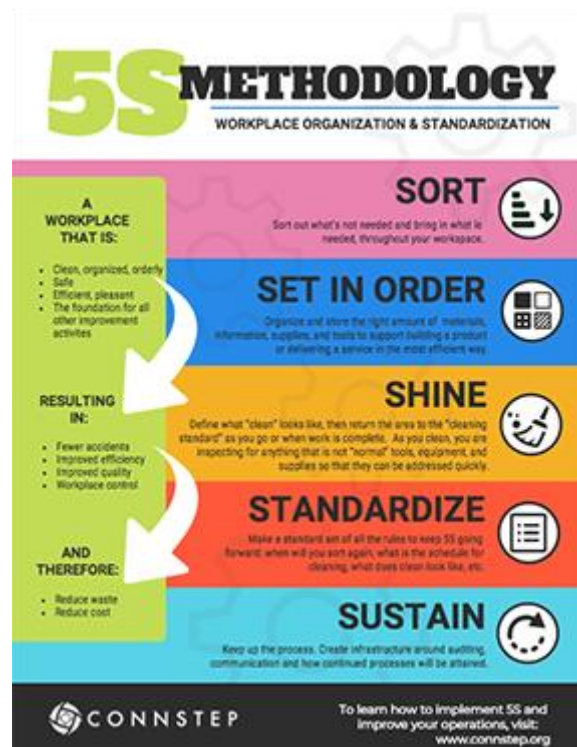


Figure 2.7 5S Housekeeping Concept

2.16 Kaizen – Continuous Improvement Philosophy

Kaizen is a Japanese philosophy that emphasizes continuous, small-step improvements involving everyone in the organization. It focuses on simplifying processes, improving each step, and building a strong culture of discipline and learning.





Figure 2.8 Kaizen Continuous Improvement